

Lecture-14 [11 September 2026]

Recap: Least squares policy evaluation (LSPE) for approximating $v_{\omega, \lambda}^{\phi}$ for a fixed ϕ

Linear architectures:

- Initialise: Specify $\underline{\beta} \in \mathbb{R}^{I+1}$, $\epsilon > 0$, $\Delta > \epsilon$.
- Iterate: While $\Delta \geq \epsilon$:
 - $\underline{v}' \leftarrow B \underline{\beta}$ ← matrix of size $M \times (I+1)$
 - $\underline{v} \leftarrow L_{\phi} \underline{v}' = r_{\phi} + \lambda P_{\phi} \underline{v}'$
 - $\underline{\beta}' \leftarrow \Gamma \underline{v}$ ← $(B^T B)^{-1} B^T$ $\underline{\beta}' \leftarrow \underset{\underline{x}}{\operatorname{argmin}} \| \underline{v} - B \underline{x} \|_2^2$
 $= \underset{\underline{x}}{\operatorname{argmin}} \sum_{s \in \mathcal{S}} (v(s) - B \underline{x}(s))^2$
 - $\Delta \leftarrow \| \underline{\beta}' - \underline{\beta} \|_2$
 - $\underline{\beta} \leftarrow \underline{\beta}'$
- Terminate: Return $\underline{\beta}$.

- Any two successive iterates $\underline{\beta}, \underline{\beta}'$ generated by LSPE satisfy

$$\underline{\beta}' = \underbrace{\Gamma r_{\phi} + \lambda \Gamma P_{\phi} B \underline{\beta}}_{g(\underline{\beta})}, \quad \Gamma = (B^T B)^{-1} B^T.$$

If $g(\cdot)$ is a contraction, then a non-stopping version of LSPE converges to the fixed point $\hat{\underline{\beta}}^*$ of $g(\cdot)$ given by

$$\hat{\underline{\beta}}^* = g(\hat{\underline{\beta}}^*) \iff \hat{\underline{\beta}}^* = (I - \lambda \Gamma P_{\phi} B)^{-1} \Gamma r_{\phi}.$$

- Non-linear architectures

- Initialise: Specify $\underline{\beta}$, $\epsilon > 0$, $\Delta > \epsilon$.
- Iterate: While $\Delta \geq \epsilon$:
 - $\underline{v}' \leftarrow \underline{f}_{\underline{\beta}}$ $v'(s) = f_{\underline{\beta}}(s) = f_{\underline{\beta}}(b_0(s), \dots, b_I(s))$
 - $\underline{v} \leftarrow L_{\phi} \underline{v}' = r_{\phi} + \lambda P_{\phi} \underline{v}'$
 - $\underline{\beta}' \in \underset{\underline{\theta}}{\operatorname{argmin}} \| \underline{f}_{\underline{\theta}} - \underline{v} \|_2^2$
 - $\Delta \leftarrow \| \underline{\beta}' - \underline{\beta} \|_2$
 - $\underline{\beta} \leftarrow \underline{\beta}'$

- Terminate: Return $\underline{\beta}$.

- A neural network-based implementation of LSPE entails training the network to generate new weights $\underline{\beta}'$ in step 3 of iterate loop corresponding to the target $\underline{v}$ from its previous step.

Weighted Least Squares

For a symmetric, positive definite matrix A of size $|S| \times |S|$, and vectors $\underline{x}, \underline{y} \in \mathbb{R}^{|S|}$, let

$$\|\underline{x}\|_A := \sqrt{\underline{x}^T A \underline{x}}, \quad \langle \underline{x}, \underline{y} \rangle_A := \underline{x}^T A \underline{y}.$$

The quantity $\|\underline{x}\|_A$ is called the Mahalanobis norm of $\underline{x}$ with respect to A .

Lemma: $\|\cdot\|_A$ is a norm on $\mathbb{R}^{|S|}$ for any symmetric, positive definite matrix A .

Proof of Lemma:

We need to show:

① $\|\underline{x}\|_A \geq 0$, with equality if and only if $\underline{x} = \underline{0}$.

② $\|\alpha \underline{x}\|_A = |\alpha| \|\underline{x}\|_A \quad \forall \alpha \in \mathbb{R}, \underline{x} \in \mathbb{R}^{|S|}$.

③ $\|\underline{x} + \underline{y}\|_A \leq \|\underline{x}\|_A + \|\underline{y}\|_A \quad \forall \underline{x}, \underline{y} \in \mathbb{R}^{|S|}$.

Part ① follows from the fact that A is positive definite.

Part ② follows by noting that

$$\sqrt{(\alpha \underline{x})^T A (\alpha \underline{x})} = \sqrt{\alpha^2 \underline{x}^T A \underline{x}} = |\alpha| \sqrt{\underline{x}^T A \underline{x}} = |\alpha| \|\underline{x}\|_A.$$

To prove part ③, note that

$$\begin{aligned} \|\underline{x} + \underline{y}\|_A^2 &= (\underline{x} + \underline{y})^T A (\underline{x} + \underline{y}) \\ &= \underline{x}^T A \underline{x} + 2 \underline{x}^T A \underline{y} + \underline{y}^T A \underline{y}. \end{aligned}$$

We now claim that $|\underline{x}^T A \underline{y}| \leq \sqrt{\underline{x}^T A \underline{x}} \cdot \sqrt{\underline{y}^T A \underline{y}}$. Indeed, for any $t \in \mathbb{R}$,

$$\begin{aligned} \|\underline{x} - t \underline{y}\|_A^2 &= (\underline{x} - t \underline{y})^T A (\underline{x} - t \underline{y}) \\ &= \underline{x}^T A \underline{x} - 2t \underline{x}^T A \underline{y} + t^2 \underline{y}^T A \underline{y} \end{aligned}$$

Noting that the quadratic expression on the RHS is strictly convex in t , its minimum is achieved for

$$t^* = \underline{x}^T A \underline{y} / \underline{y}^T A \underline{y}.$$

and

$$\underline{x}^T A \underline{x} - 2t^* \underline{x}^T A \underline{y} + (t^*)^2 \underline{y}^T A \underline{y} = \|\underline{x} - t^* \underline{y}\|_A^2 \geq 0$$

$$\Leftrightarrow (\underline{x}^T A \underline{y})^2 \leq (\underline{x}^T A \underline{x}) (\underline{y}^T A \underline{y})$$

$$\Leftrightarrow |\underline{x}^T A \underline{y}| \leq \|\underline{x}\|_A \|\underline{y}\|_A.$$

Hence,

$$\begin{aligned} \|\underline{x} + \underline{y}\|_A^2 &\leq \|\underline{x}\|_A^2 + 2\|\underline{x}\|_A \|\underline{y}\|_A + \|\underline{y}\|_A^2 \\ &= (\|\underline{x}\|_A + \|\underline{y}\|_A)^2, \end{aligned}$$

from which part (3) follows.

Remark: If A is symmetric & positive definite, then

$$A = U \Sigma U^T, \quad A^{1/2} = U \Sigma^{1/2} U^T.$$

↪ diagonal

In this case, $\|\underline{x}\|_A = \|A^{1/2} \underline{x}\|_2.$

Given a diagonal weight matrix $W = \text{diag}(w_1, \dots, w_{|S|})$, with $w_i \geq 0 \ \forall i$,
Consider the weighted least squares problem

$$\min_{\underline{\beta} \in \mathbb{R}^{I+1}} \|\underline{B}\underline{\beta} - \underline{v}_{\omega, \lambda}^\phi\|_W^2.$$

The optimal solution $\hat{\underline{\beta}}_{\text{WLS}}$ to the above problem is given by

$$\hat{\underline{\beta}}_{\text{WLS}} = \underbrace{(\underline{B}^T W \underline{B})^{-1}}_{=: \underline{\Gamma}_W} \underline{B}^T W \underline{v}_{\omega, \lambda}^\phi = \underline{\Gamma}_W \underline{v}_{\omega, \lambda}^\phi$$

and the optimal approximation is

$$\hat{\underline{v}}_{\text{WLS}} = \underline{B} \hat{\underline{\beta}}_{\text{WLS}} = \underline{B} \underbrace{(\underline{B}^T W \underline{B})^{-1}}_{=: \underline{\Pi}_W = \underline{B} \underline{\Gamma}_W} \underline{B}^T W \underline{v}_{\omega, \lambda}^\phi = \underline{\Pi}_W \underline{v}_{\omega, \lambda}^\phi.$$

$$\begin{aligned} \underline{\Pi}_W^2 &= \underline{\Pi}_W \\ \underline{\Pi}_W W &= W \underline{\Pi}_W^T \end{aligned}$$

Theorem: Fix $\phi: \mathcal{S} \rightarrow \mathcal{A}$. Let $\mathcal{S} = \{s_1, \dots, s_m\}$. Suppose P_ϕ has a unique

left eigenvector $\underline{\xi} = [\xi_1, \dots, \xi_m]^T$ satisfying

$$\underline{\xi}^T = \underline{\xi}^T P_\phi, \quad \xi_i > 0 \ \forall i, \quad \sum_i \xi_i = 1.$$

Let $B_{m \times I+1}$ have full column rank, and let $W = \text{diag}(\xi_1, \dots, \xi_m)$.

Then, $\Pi_W L_\phi: \mathbb{R}^{|S|} \rightarrow \mathbb{R}^{|S|}$ is a contraction with modulus λ under the weighted norm $\|\cdot\|_W$, i.e.,

$$\|\Pi_W L_\phi \underline{v} - \Pi_W L_\phi \underline{u}\|_W \leq \lambda \|\underline{v} - \underline{u}\|_W \quad \forall \underline{u}, \underline{v} \in \mathbb{R}^{|S|}.$$

Proof of Theorem

- Step 1: Show that $\|P_\phi \underline{v}\|_W \leq \|\underline{v}\|_W \quad \forall \underline{v} \in \mathbb{R}^{|S|}$.

To show this, we note that

$$\begin{aligned} \|P_\phi \underline{v}\|_W^2 &= \sum_{i=1}^m \xi_i \left(\sum_{j=1}^m P(s_j | s_i, \phi(s_i)) v(s_j) \right)^2 \\ &= \sum_{i=1}^m \xi_i \left(\sum_{j=1}^m \sqrt{P(s_j | s_i, \phi(s_i))} \sqrt{P(s_j | s_i, \phi(s_i))} v(s_j) \right)^2 \end{aligned}$$

$$\stackrel{\text{Cauchy}}{\leq} \sum_{i=1}^m \xi_i \left(\sum_{j=1}^m P(s_j | s_i, \phi(s_i)) \right) \left(\sum_{j=1}^m P(s_j | s_i, \phi(s_i)) v^2(s_j) \right) \stackrel{\text{Schwarz}}{}$$

$$= \sum_{i=1}^m \xi_i \sum_{j=1}^m P(s_j | s_i, \phi(s_i)) v^2(s_j)$$

$$= \sum_{j=1}^m \sum_{i=1}^m \xi_i \underbrace{P(s_j | s_i, \phi(s_i))}_{P_\phi(s_j | s_i)} \cdot v^2(s_j)$$

$$= \sum_{j=1}^m \underbrace{\sum_{i=1}^m \xi_i P_\phi(s_j | s_i)}_{\xi_j} \cdot v^2(s_j) \quad \underline{\xi} = \underline{\xi} P_\phi$$

$$= \sum_{j=1}^m \xi_j v^2(s_j)$$

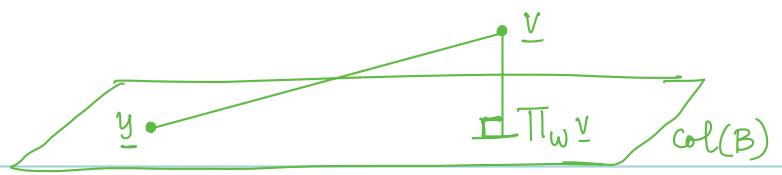
$$= \|\underline{v}\|_W^2.$$

Step 2: We now prove that

$$\|\Pi_W \underline{v} - \Pi_W \underline{u}\|_W \leq \|\underline{v} - \underline{u}\|_W \quad \forall \underline{u}, \underline{v} \in \mathbb{R}^{|S|}.$$

To see this, observe that for any $\underline{y} \in \text{Col}(B)$

$$\langle \underline{y}, \underline{v} - \Pi_W \underline{v} \rangle_W = 0. \quad (\text{Exercise})$$



In particular,

$$\langle \Pi_w \underline{v}, \underline{v} - \Pi_w \underline{v} \rangle_w = 0.$$

Hence, we have

$$\begin{aligned} \|\underline{v}\|_w^2 &= \|\underline{v} - \Pi_w \underline{v} + \Pi_w \underline{v}\|_w^2 \\ &= \|\underline{v} - \Pi_w \underline{v}\|_w^2 + \|\Pi_w \underline{v}\|_w^2 \\ &\geq \|\Pi_w \underline{v}\|_w^2. \end{aligned}$$

Applying the above result to $\underline{v} \leftarrow \underline{v} - \underline{u}$, and noting that

$\Pi_w = B(B^T W B)^{-1} B^T W$ is linear, we get

$$\begin{aligned} \|\Pi_w \underline{v} - \Pi_w \underline{u}\|_w &\leq \|\Pi_w (\underline{v} - \underline{u})\|_w \\ &\leq \|\underline{v} - \underline{u}\|_w. \end{aligned}$$

Step 3: Show that L_ϕ is a contraction with modulus λ under $\|\cdot\|_w$.

We have

$$\begin{aligned} \|L_\phi \underline{v} - L_\phi \underline{u}\|_w &= \|\pi_\phi + \lambda P_\phi \underline{v} - \pi_\phi - \lambda P_\phi \underline{u}\|_w \\ &= \lambda \|P_\phi (\underline{v} - \underline{u})\|_w \\ &\leq \lambda \|\underline{v} - \underline{u}\|_w \quad (\text{Step 1}). \end{aligned}$$

Step 4: Combine the results of step 2 and step 3:

$$\begin{aligned} \|\Pi_w L_\phi \underline{v} - \Pi_w L_\phi \underline{u}\|_w &= \|\Pi_w (L_\phi \underline{v} - L_\phi \underline{u})\|_w \\ &\leq \|L_\phi \underline{v} - L_\phi \underline{u}\|_w \quad (\text{Step 2}) \\ &\leq \lambda \|\underline{v} - \underline{u}\|_w \quad (\text{Step 3}). \end{aligned}$$

A Weighted Least Squares Version of LSPE Under Linear Architectures

The ordinary least squares-based LSPE algorithm may not converge always. The below example is a case-in-point.

Example [to demonstrate that $(I - \lambda \Gamma P_\phi B)$ may not be invertible]

Let $S = \{s_1, s_2\}$, $\lambda = \frac{5}{5.4}$, $P_\phi = \begin{bmatrix} 0.2 & 0.8 \\ 0.2 & 0.8 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Then, $\Gamma = (B^T B)^{-1} B^T = \frac{1}{5} \begin{bmatrix} 1 & 2 \end{bmatrix}$, $\Gamma P_\phi B = \frac{5.4}{5}$

and therefore

$$\lambda \Gamma P_\phi B = \frac{5}{5.4} \cdot \frac{1}{5} \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 0.2 & 0.8 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1.$$

Thus, $I - \lambda \Gamma P_\phi B$ (in this a 1×1 matrix) is not invertible!

In this case, we note that the successive iterates β_n, β_{n+1} of LSPE satisfy

$$\begin{aligned} \beta_{n+1} &= \Gamma r_\phi + \lambda \Gamma P_\phi B \beta_n \\ &= \Gamma r_\phi + \beta_n. \end{aligned}$$

Thus, if r_ϕ is such that $\Gamma r_\phi \neq 0$, then $\{\beta_n\}_{n \geq 0}$ does not converge.

If r_ϕ is such that $\Gamma r_\phi = 0$, then $\{\beta_n\}_{n \geq 0} = \{\beta_0, \beta_0, \dots\}$, which is not unique and changes with the initial choice β_0 .

A Weighted Least Squares Variant of LSPE

- Initialize: Specify $\beta_- \in \mathbb{R}^{I+1}$, $\varepsilon > 0$, $\Delta > \varepsilon$, feature matrix B .
- Iterate: While $\Delta \geq \varepsilon$:
 - $\underline{v}' \leftarrow B \beta_-$
 - $\underline{v} \leftarrow L_\phi \underline{v}' = r_\phi + \lambda P_\phi \underline{v}'$
 - $\beta'_- \leftarrow \Gamma_w \underline{v}$ $\beta'_- \leftarrow \argmin_x \|\underline{v} - Bx\|_w^2$
 - $\Delta \leftarrow \|\beta'_- - \beta_-\|_w$ (one can also use $\|\beta'_- - \beta_-\|_2$ here)
 - $\beta_- \leftarrow \beta'_-$
- Terminate: Return β_- .

Theorem: Suppose that there exists $\underline{\xi} = (\xi_1, \dots, \xi_M)$ satisfying

$$\xi_i > 0 \quad \forall i \in \{1, \dots, M\}, \quad \underline{\xi} = \underline{\xi} P_\phi.$$

Also, suppose that the feature matrix B has full column rank. Then,

- ① The sequence $\{\beta_n\}_{n \geq 0}$ generated by the WLS-based LSPE algorithm with $W = \text{diag}(\xi_1, \dots, \xi_M)$ converges in $\|\cdot\|_M$, where

$$M = B^T W B.$$

- ② The matrix $I - \lambda \Gamma_w P_\phi B$ is invertible, and

$$\beta_n \rightarrow \beta^* = (I - \lambda \Gamma_w P_\phi B)^{-1} \Gamma_w x_\phi.$$

Proof of Theorem: Under the hypotheses of the above Theorem, we know that $\Pi_w L_\phi$ is a contraction under $\|\cdot\|_w$ with modulus λ .

Notice that the iterates β_n, β_{n+1} of WLS-based LSPE are related as

$$\beta_{n+1} = \underbrace{\Gamma_w x_\phi + \lambda \Gamma_w P_\phi B \beta_n}_{g_w(\beta)} \quad , \quad \Gamma_w = (B^T W B)^{-1} B^T W.$$

Also, notice that

$$\|B\beta\|_w = \sqrt{\beta^T B^T W B \beta} = \sqrt{\beta^T M \beta} = \|\beta\|_M.$$

Importantly:

$$W \text{ symmetric, PD} \Rightarrow M \text{ symmetric, PD}$$

$$B \text{ full rank}$$

$$\Rightarrow \|\cdot\|_M \text{ is a norm on the space of } \beta \text{ vectors}$$

Then, we have

$$\begin{aligned} \|g_w(\beta) - g_w(\tilde{\beta})\|_M &= \|B(g_w(\beta) - g_w(\tilde{\beta}))\|_w \\ &= \|\Pi_w L_\phi(B\beta) - \Pi_w L_\phi(B\tilde{\beta})\|_w \\ &\leq \lambda \|B\beta - B\tilde{\beta}\|_w \\ &= \lambda \|\beta - \tilde{\beta}\|_M. \end{aligned}$$

This proves that g_w is a contraction under $\|\cdot\|_M$. Because

the space of $\underline{\beta}$ vectors is complete under $\|\cdot\|_m$ (all norms on the space of $\underline{\beta}$ vectors are equivalent), we get that

$$\underline{\beta}_n \rightarrow \underline{\beta}^* = g_w(\underline{\beta}^*).$$

Now,

$$\begin{aligned} \underline{\beta}^* = g_w(\underline{\beta}^*) &\Leftrightarrow \underline{\beta}^* = \Gamma_w \underline{x}_\phi + \lambda \Gamma_w P_\phi B \underline{\beta}^* \\ &\Leftrightarrow (\mathbf{I} - \lambda \Gamma_w P_\phi B) \underline{\beta}^* = \Gamma_w \underline{x}_\phi. \end{aligned}$$

We now show that $(\mathbf{I} - \lambda \Gamma_w P_\phi B)$ is invertible. Observe that

$$\|g_w(\underline{\beta}) - g_w(\tilde{\underline{\beta}})\|_m = \|\lambda \Gamma_w P_\phi B (\underline{\beta} - \tilde{\underline{\beta}})\|_m$$

$$\leq \lambda \|\underline{\beta} - \tilde{\underline{\beta}}\|_m \quad (\text{as established earlier}). \quad (1)$$

Suppose that $(\mathbf{I} - \lambda \Gamma_w P_\phi B) \hat{\underline{\beta}} = \underline{0}$ for some $\hat{\underline{\beta}}$. Then,

$$(\mathbf{I} - \lambda \Gamma_w P_\phi B) \hat{\underline{\beta}} = \underline{0} \Leftrightarrow \hat{\underline{\beta}} = \lambda \Gamma_w P_\phi B \hat{\underline{\beta}}$$

Using (1) with $\tilde{\underline{\beta}} \leftarrow \hat{\underline{\beta}}$ and $\underline{\beta} \leftarrow \underline{0}$, we get

$$\|\hat{\underline{\beta}}\|_m = \|\lambda \Gamma_w P_\phi B \hat{\underline{\beta}}\|_m \leq \lambda \|\hat{\underline{\beta}}\|_m \Rightarrow \|\hat{\underline{\beta}}\|_m = 0 \Rightarrow \underline{\beta} = \underline{0}.$$

Thus, $\mathbf{I} - \lambda \Gamma_w P_\phi B$ is invertible, and part (2) follows.

Revisiting the example from before:

$$\text{Let } S = \{s_1, s_2\}, \quad \lambda = \frac{5}{5.4}, \quad P_\phi = \begin{bmatrix} 0.2 & 0.8 \\ 0.2 & 0.8 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

$$\text{Here, } \underline{\xi} = \begin{bmatrix} 1/5 \\ 4/5 \end{bmatrix}, \quad W = \begin{pmatrix} 1/5 & 0 \\ 0 & 4/5 \end{pmatrix}, \quad B^T W B = \frac{17}{5},$$

$$\Gamma_w = (B^T W B)^{-1} B^T W = \frac{1}{17} \begin{bmatrix} 1 & 8 \end{bmatrix}.$$

Therefore,

$$\Pi_w = \frac{1}{17} \begin{pmatrix} 1 & 8 \\ 2 & 16 \end{pmatrix}, \quad \lambda \Gamma_w P_\phi B = \frac{15}{17} < 1.$$

For this example, the WLS-based LSPE sequence $\{\underline{\beta}_n\}_{n \geq 0}$ satisfies

$$\underline{\beta}_{n+1} = \Gamma_w \underline{x}_\phi + \frac{15}{17} \underline{\beta}_n, \quad n \geq 0.$$

$$= \sum_{i=1}^M \xi_i \sum_{j=1}^M p_{\phi}(s_j | s_i) b(s_i) b(s_j)^T$$

$$= B^T W P_{\phi} B = C_{\phi},$$

$$\frac{1}{n} \sum_{t=1}^n \alpha_t R_{t+1} = \frac{1}{n} \sum_{t=1}^n b(s_t) r_{\phi}(s_t)$$

$$\xrightarrow[n.s.]{n \rightarrow \infty} \mathbb{E}_{S \sim \xi} [b(s) r_{\phi}(s)]$$

$$= \sum_{i=1}^M \xi_i b(s_i) r_{\phi}(s_i)$$

$$= B^T W r_{\phi} = \underline{d}_{\phi}.$$

$$\underbrace{\frac{1}{n} \sum_{t=1}^n b(s_t) b(s_t)^T}_{\hat{M}_n} \xrightarrow[n.s.]{n \rightarrow \infty} \underbrace{B^T W B}_M, \quad \underbrace{\frac{1}{n} \sum_{t=1}^n b(s_t) b(s_{t+1})^T}_{\hat{C}_n} \xrightarrow[n.s.]{n \rightarrow \infty} \underbrace{B^T W P_{\phi} B}_{C_{\phi}},$$

$$\underbrace{\frac{1}{n} \sum_{t=1}^n b(s_t) r_{\phi}(s_t)}_{\hat{\underline{d}}_n} \xrightarrow[n.s.]{n \rightarrow \infty} \underbrace{B^T W r_{\phi}}_{\underline{d}_{\phi}}$$

Run the policy $\pi = \{\phi, \phi, \dots\}$ till a sufficiently large time N for which

$$\|\hat{M}_N - \hat{M}_{N-1}\| < \varepsilon, \quad \|\hat{C}_N - \hat{C}_{N-1}\| < \varepsilon, \quad \|\hat{\underline{d}}_N - \hat{\underline{d}}_{N-1}\|_M < \varepsilon.$$

Then, in WLS-based LSPE, replace the step

$$\underline{\beta}' \leftarrow \Gamma_W \underline{\beta} = M^{-1} (\underline{d}_{\phi} + \lambda C_{\phi} \underline{\beta})$$

with the following:

$$\underline{\beta}' \leftarrow (\hat{M}_N)^{-1} (\hat{\underline{d}}_N + \lambda \hat{C}_N \underline{\beta})$$

Remarks :

- ① The (OLS-based) LSPE algorithm is not entirely useless. In some cases, it does not converge, but in many cases, it does converge. Notably,

largest absolute eigenvalue

$$\rho(\lambda \Gamma P_{\phi} B) < 1 \iff \{P_n\}_{n \geq 0} \text{ arising from LSPE converges to } \hat{P}^* = (I - \lambda \Gamma P_{\phi} B)^{-1} \Gamma r_{\phi}.$$

- ② In some problems, OLS-based LSPE iterates $\{P_n\}_{n \geq 0}$ will not converge. In such cases, one can try out simulation-based LSPE.
- ③ For the simulation-based LSPE, the quantities $\hat{M}_N$, $\hat{C}_N$, $\hat{d}_N$ can be computed by running ϕ offline and generating the MDP trajectory.
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